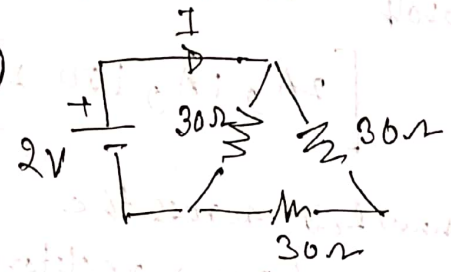
Problem of Division of current in '2' & '3' branches. Combination of cells. on Voltmeter / Ammeter etc.

(1) A long uniform wire of resistance 9Ω is bent into a circle. Two points of the wire at one-third of the circumference apart are joined with a cell of emf 1.5V and internal resistance 0.5Ω. Calculate currents through all parts of circuit

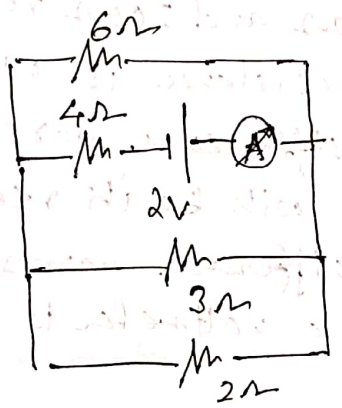
[0.6A, 0.4A, 0.2A]

(2)



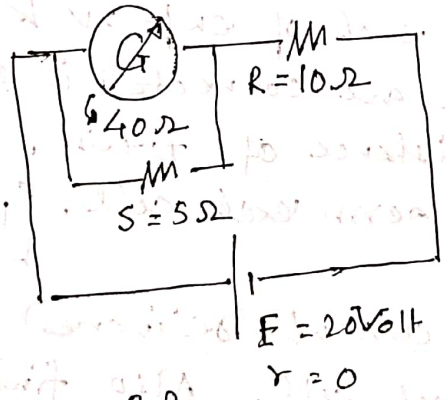
Find I ? $\left(\frac{1}{10} A\right)$

(3)



What is the reading of ammeter in this circuit? (0.4A)

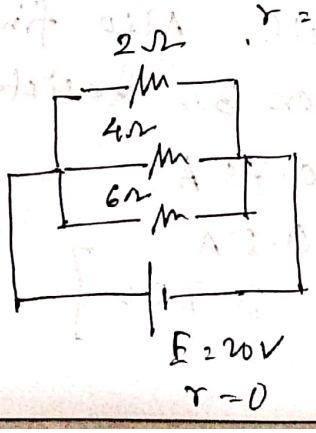
(4)



Find current through the galvanometer.

[2/13 A]

(5)



Find current through each resistor

(10A, 5A, 4A).

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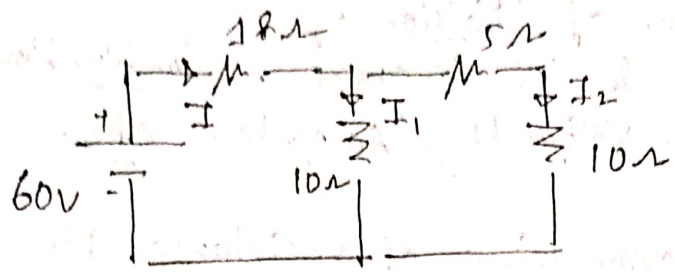
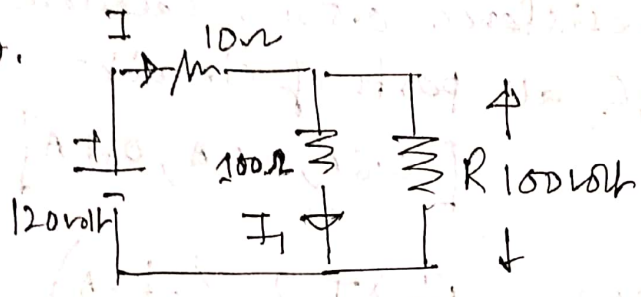
6.



Find I, I_1, I_2 & voltage drop across 18Ω

$\left[I = \frac{5}{2}A,\ I_1 = \frac{3}{2}A,\ I_2 = 1A,\ 45V\right]$

7.



find I, I_1 & R = ?

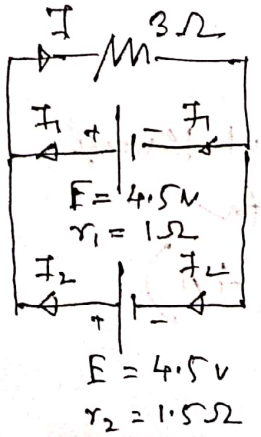
$[2A,\ 1A,\ 100\Omega]$

8. A 84 volt battery is connected across the series combination of 100Ω and 200Ω resistors. A voltmeter of 400Ω resistance is used to measure the potential difference across the 100Ω resistor. What will be its reading? What is the potential difference across the 100Ω resistor before the voltmeter is connected?

[24V, 28V]

9. Three identical cells each of emf 2V & internal resistance 0.2Ω are connected in series to an external resistance of 7.4Ω. Find potential difference across each cell. [1.85volt]

10.



Find current delivered by each cell. Also find p.d across 3Ω resistance

$[I_1 = 0.75A$
$I_2 = 0.5A$
$V = 3.75V]$

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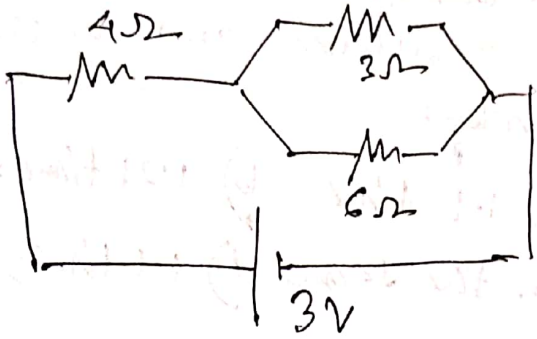
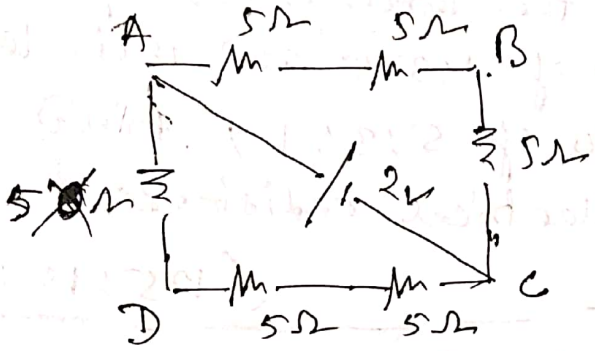
17.



Find P.D across 3Ω resistance?

[1V]

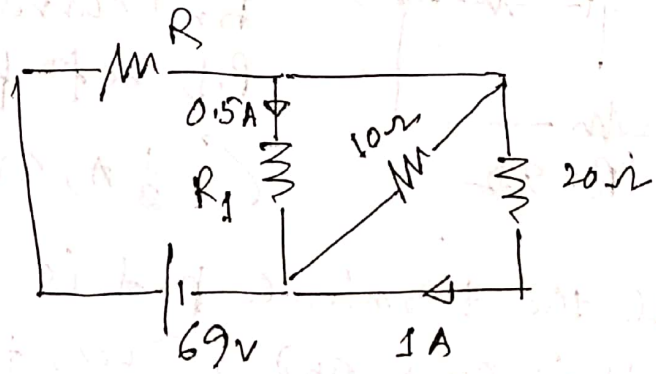
18.



Find PD between pts A & B

[4/3 V]

19.



Find R, R_1

[14Ω, 40Ω]

20. Two wires of same length and cross sections but of resistivities ρ_1 and ρ_2 are connected in series. Find equivalent resistivity of this series combination. $\left(\frac{\rho_1+\rho_2}{2}\right)$

21. '12' identical cells each of emf '2V' (some of them are wrongly connected) are in series across a resistance. If current in the circuit is 2A and total resistance of circuit (resistance of external resistor + net resistance of cells) is 10Ω, then find the number of cells connected wrongly? [1]

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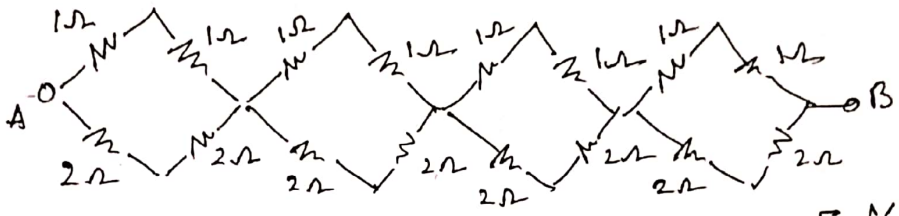
(11) Three identical cells each of emf 2V and internal resistance 1Ω are connected in series to form a battery. The battery is then connected to parallel combination of two identical resistors, each of resistance 6Ω. Find current delivered by the battery. [1A]

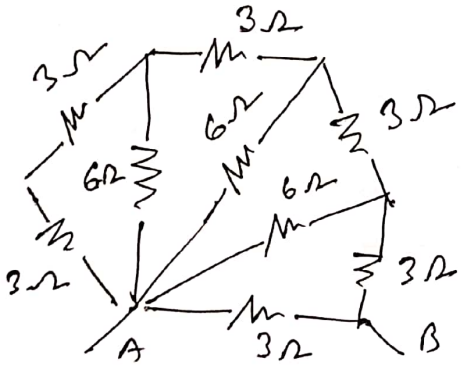
(12) Two wires of same metal have same length, but their cross-sections are in the ratio 3:1. They are joined in series. The resistance of the thicker wire is 10Ω. Find equivalent resistance of the combination. [40Ω]

(13)



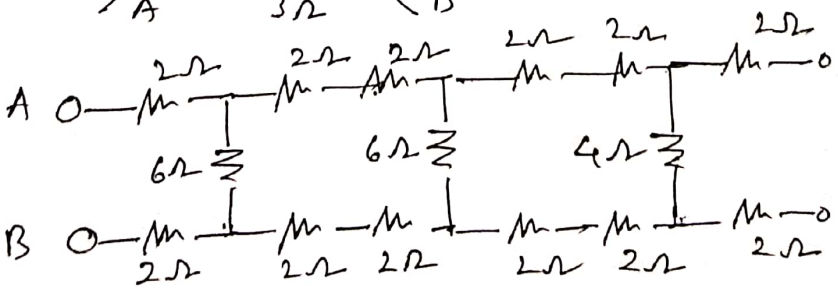
Find equivalent resistance betn A & B [$\frac{16}{3}$Ω]

(14)



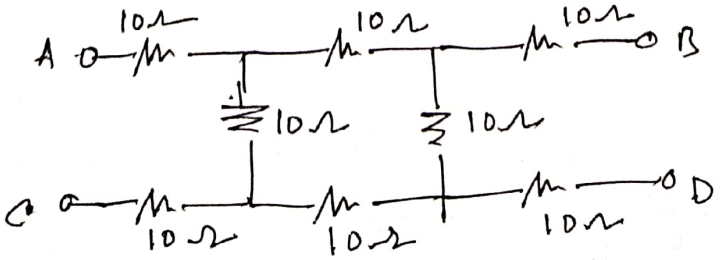
Find equivalent resistance betn A & B [2Ω]

(15)



Find RAB [8Ω]

(16)



Find equivalent resistance betn A & D. [30Ω]

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